

Project wireless

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Ping ubuntu from kali

```
(kali@kali)-[~]
$ ping -c 4 192.168.174.130
PING 192.168.174.130 (192.168.174.130) 56(84) bytes of data.
64 bytes from 192.168.174.130: icmp_seq=1 ttl=64 time=0.908 ms
64 bytes from 192.168.174.130: icmp_seq=2 ttl=64 time=0.809 ms
64 bytes from 192.168.174.130: icmp_seq=3 ttl=64 time=1.01 ms
64 bytes from 192.168.174.130: icmp_seq=4 ttl=64 time=0.887 ms

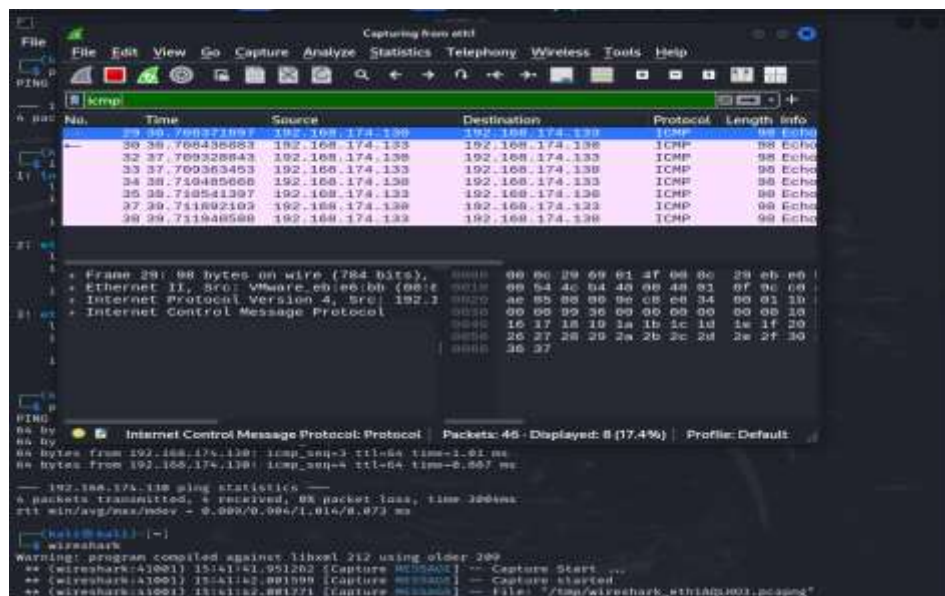
--- 192.168.174.130 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.809/0.904/1.014/0.073 ms
```

Ping kali from ubuntu

```
cyberarena@ubuntu:~$ ping -c 4 192.168.174.133
PING 192.168.174.133 (192.168.174.133) 56(84) bytes of data.
64 bytes from 192.168.174.133: icmp_seq=1 ttl=64 time=0.815 ms
64 bytes from 192.168.174.133: icmp_seq=2 ttl=64 time=0.757 ms
64 bytes from 192.168.174.133: icmp_seq=3 ttl=64 time=0.687 ms
64 bytes from 192.168.174.133: icmp_seq=4 ttl=64 time=0.621 ms

--- 192.168.174.133 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.621/0.720/0.815/0.072 ms
cyberarena@ubuntu:~$
```

Icmp in wireshark



Week 2

```
(kali@kali)~$ nmap -sn 192.168.174.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:44 EDT
Nmap scan report for 192.168.174.1
Host is up (0.00065s latency).
MAC Address: 00:50:56:C0:00:01 (VMware)
Nmap scan report for 192.168.174.130
Host is up (0.00032s latency).
MAC Address: 00:0C:29:EB:E6:8B (VMware)
Nmap scan report for 192.168.174.254
Host is up (0.00026s latency).
MAC Address: 00:50:56:F7:6B:7A (VMware)
Nmap scan report for 192.168.174.133
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 15.14 seconds
```

Commented [s1]: Sn for host discovery

192.168.174.130 → Ubuntu target

192.168.174.133 → Kali attacker

```
(kali@kali)~$ sudo nmap -sS -p 1-1000 192.168.174.130
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:46 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00054s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 00:0C:29:EB:E6:BB (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.05 seconds
```

Commented [s2]: -Ss stealth scan

This performs a SYN scan against the Ubuntu target for ports 1–1000.

```
(kali@kali)~$ sudo nmap -sV 192.168.174.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:47 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00036s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache/2.4.18 (Ubuntu)
3000/tcp   open  http     Node.js Express framework
8080/tcp   open  http     Apache/2.4.18 (Ubuntu)
MAC Address: 00:0C:29:EB:E6:BB (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 17.64 seconds
```

Commented [s3]: -Sv for service version

```

kali@kali:~$ sudo nmap -O 192.168.174.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:48 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00081s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
3000/tcp   open  ppp
8080/tcp   open  http-proxy
MAC Address: 00:0C:29:EB:E6:BB (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.8
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.21 seconds

```

Commented [s4]: -O for operating system

```

kali@kali:~$ sudo nmap -A 192.168.174.130 -oN nmap_week2_report.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:50 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00049s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   256 afd4:4e:53:8b:13:ce:73:8a:6e:fe:eb:8c:78:96:a8 (ECDSA)
|_ 256 8d:02:aa:2c:ee:40:fd:5d:96:9c:f7:36:7c:b8:05:03 (ED25519)
80/tcp    open  http     Apache/2.4.18 (Ubuntu)
|_ http-title: Nextcloud
|_ http-server-header: Apache
|_ http-robots.txt: 1 disallowed entry
|_
3000/tcp   open  http     Node.js Express Framework
|_ http-title: Rocket.Chat
|_ http-robots.txt: 1 disallowed entry
|_
8080/tcp   open  http     Apache Tomcat/9.0.112
|_ http-title: HTTP Status 404 \xE2\x80\x93 Not Found
MAC Address: 00:0C:29:EB:E6:BB (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.8
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   0.50 ms  192.168.174.130

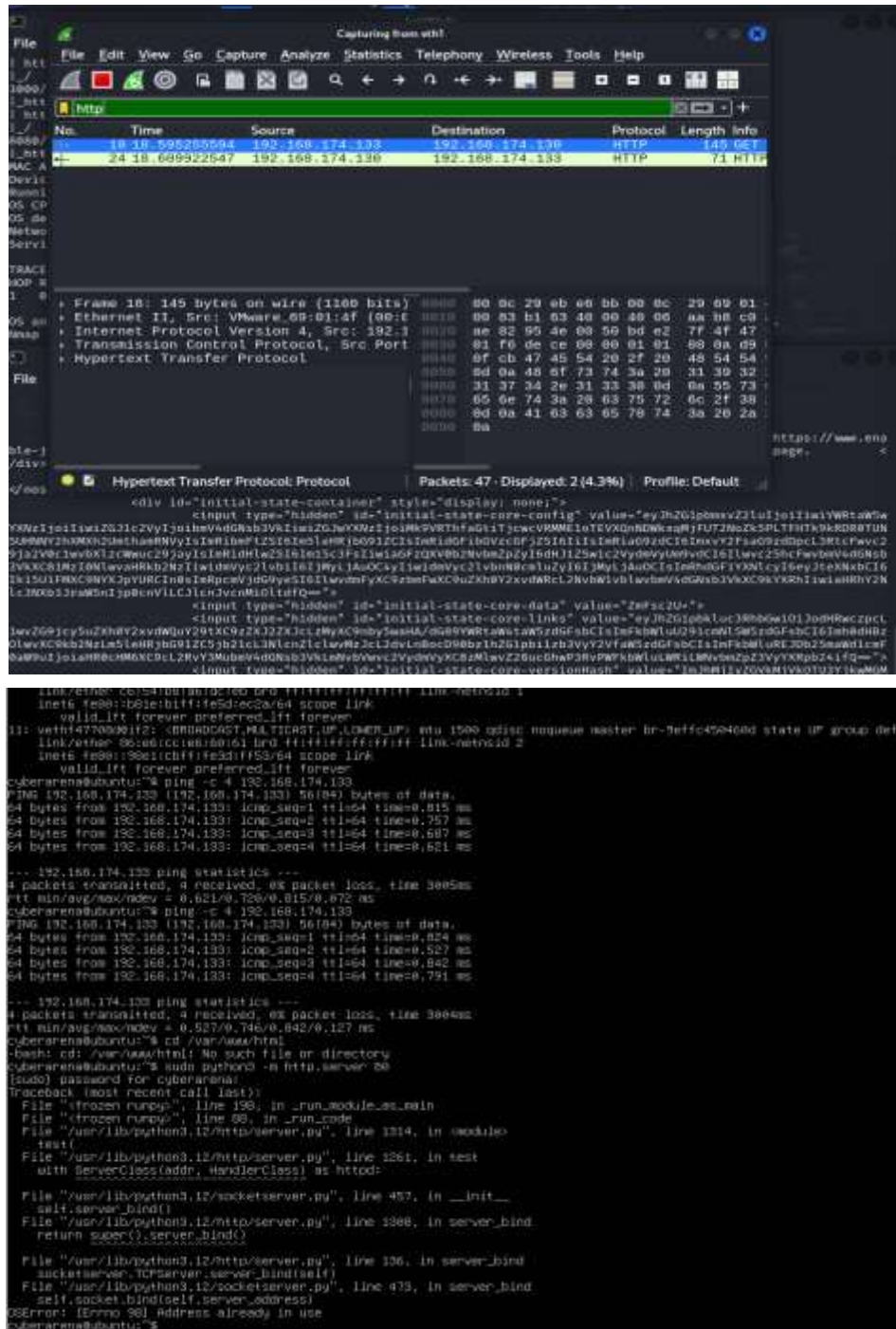
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.82 seconds

kali@kali:~$ ls
@_Number_Systems.pptx      Flag83.rar              risestrings.txt
active_contracts_by_vendor_name_excluded.xls  hash.txt                80x80totheChallengehaveData864.jpg
Animation.html             hound                   routersploit.log
compress2                  informant.oct            sample_resume.pdf
customers.docx             lolang.py               sample_resumes.pdf
CVF1727_Roster_2028.csv   n337m333               ScriptablePointer.java
CVF1727_Roster_2028.txt   metadata_sqlite_db      spoof.cap
CVF1727_Roster_2028.xlsx  music                  TeamViewerInstall.exe
DefaultIDGMStructureHelperImpl.java          nmap_week2_report.txt  Templates

```

Commented [s5]: -A AGGRESSIVE -oN save out nmap in text file

Week 3



The image displays a Wireshark packet capture of an HTTP GET request and a terminal window showing a web server running on a Raspberry Pi.

Wireshark Packet Capture:

No.	Time	Source	Destination	Protocol	Length	Info
18	18.59525504	192.168.174.133	192.168.174.133	HTTP	145	GET
24	18.609922547	192.168.174.133	192.168.174.133	HTTP	71	HTTP

Frame 18: 145 bytes on wire (1160 bits)

- Ethernet II, Src: VMware 08:00:14:00:00:00, Dst: 192.168.174.133
- Internet Protocol Version 4, Src: 192.168.174.133, Dst: 192.168.174.133
- Transmission Control Protocol, Src Port: 54321, Dst Port: 80
- Hypertext Transfer Protocol

Frame 24: 71 bytes on wire (568 bits)

- Ethernet II, Src: VMware 08:00:14:00:00:00, Dst: 192.168.174.133
- Internet Protocol Version 4, Src: 192.168.174.133, Dst: 192.168.174.133
- Transmission Control Protocol, Src Port: 80, Dst Port: 54321
- Hypertext Transfer Protocol

Terminal Window:

```

link/ether: 08:00:14:00:00:00 link-local: ff:ff:ff:ff:ff:ff link-metric: 1
inet6 fe80::b81e:11ff:fe5d:ec2a::64 scope link
valid_lft forever preferred_lft forever
11: veth477000012: <BROADCAST,MULTICAST,UP,LOWER_UP> eno1500 qdisc noqueue master br-9effc450400d state UP group default
link/ether: 08:00:14:00:00:00 link-local: ff:ff:ff:ff:ff:ff link-metric: 2
inet6 fe80::b81e:11ff:fe5d:ec2a::64 scope link
valid_lft forever preferred_lft forever
cyberarena@ubuntu:~$ ping -c 4 192.168.174.133
PING 192.168.174.133 (192.168.174.133) 56(84) bytes of data:
64 bytes from 192.168.174.133: icmp_seq=1 ttl=64 time=0.815 ms
64 bytes from 192.168.174.133: icmp_seq=2 ttl=64 time=0.757 ms
64 bytes from 192.168.174.133: icmp_seq=3 ttl=64 time=0.687 ms
64 bytes from 192.168.174.133: icmp_seq=4 ttl=64 time=0.621 ms
--- 192.168.174.133 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/ndev = 0.621/0.726/0.815/0.672 ms
cyberarena@ubuntu:~$ ping -c 4 192.168.174.133
PING 192.168.174.133 (192.168.174.133) 56(84) bytes of data:
64 bytes from 192.168.174.133: icmp_seq=1 ttl=64 time=0.824 ms
64 bytes from 192.168.174.133: icmp_seq=2 ttl=64 time=0.527 ms
64 bytes from 192.168.174.133: icmp_seq=3 ttl=64 time=0.842 ms
64 bytes from 192.168.174.133: icmp_seq=4 ttl=64 time=0.791 ms
--- 192.168.174.133 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/ndev = 0.527/0.746/0.842/0.127 ms
cyberarena@ubuntu:~$ cd /var/www/html
-bash: cd: /var/www/html: No such file or directory
cyberarena@ubuntu:~$ sudo python3 -m http.server 80
[sudo] password for cyberarena:
Traceback (most recent call last):
  File "/usr/lib/python3.12/http/server.py", line 130, in __init__
    self.server_bind()
  File "/usr/lib/python3.12/http/server.py", line 136, in server_bind
    super().server_bind()
  File "/usr/lib/python3.12/socketserver.py", line 475, in __init__
    self.socket.bind(self.server_address)
OSError: [Errno 98] Address already in use
cyberarena@ubuntu:~$

```


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Capturing from eth0

No.	Time	Source	Destination	Protocol	Length	Info
51	33.818529106	192.168.174.133	192.168.174.133	FTP	86	Req
56	45.166473320	192.168.174.133	192.168.174.133	FTP	63	Req
66	46.167080759	192.168.174.133	192.168.174.133	FTP	100	Req
74	51.977471267	192.168.174.133	192.168.174.133	FTP	100	Req
78	55.172055111	192.168.174.133	192.168.174.133	FTP	88	Req

Frame 51: 86 bytes on wire (688 bits), 0000 00 8c 29 69 01 4f 00 8c 29 eb e6
 Ethernet II, Src: VMware_eb:c6:bb (08:00:0e:08:00:00), Dst: 08:00:0e:08:00:00
 Internet Protocol Version 4, Src: 192.168.174.133, Dst: 192.168.174.133
 Transmission Control Protocol, Src Port: 21, Dst Port: 21
 File Transfer Protocol (FTP)
 [Current working directory:]

File Actions Edit View Help

```

kali@kali:~$ cat /home/kali/.ssh_history
kali@kali:~$ ssh 192.168.174.133
Connected to 192.168.174.133.
220 (vsFTPd 3.0.5)
Name (192.168.174.133:kali): cyberarena
331 Please specify the password.
  
```

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Capturing from eth0

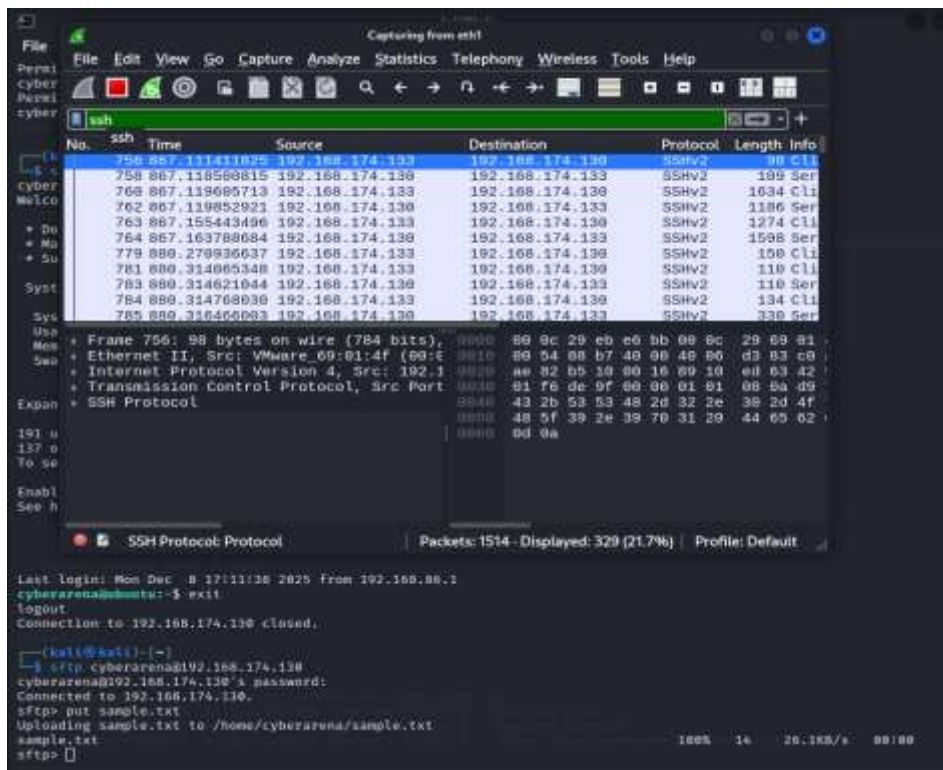
No.	Time	Source	Destination	Protocol	Length	Info
581	671.726821775	192.168.174.133	192.168.174.133	FTP	83	Req
583	671.727179147	192.168.174.133	192.168.174.133	FTP	100	Req
591	679.329281185	192.168.174.133	192.168.174.133	FTP	83	Req
595	681.932465893	192.168.174.133	192.168.174.133	FTP	88	Req
611	704.178169856	192.168.174.133	192.168.174.133	FTP	72	Req
613	704.178470042	192.168.174.133	192.168.174.133	FTP	104	Req
615	704.178548804	192.168.174.133	192.168.174.133	FTP	72	Req
616	704.178874716	192.168.174.133	192.168.174.133	FTP	104	Req
617	704.179002301	192.168.174.133	192.168.174.133	FTP	96	Req
618	704.179256769	192.168.174.133	192.168.174.133	FTP	104	Req

Frame 51: 86 bytes on wire (688 bits), 0000 00 8c 29 69 01 4f 00 8c 29 eb e6
 Ethernet II, Src: VMware_eb:c6:bb (08:00:0e:08:00:00), Dst: 08:00:0e:08:00:00
 Internet Protocol Version 4, Src: 192.168.174.133, Dst: 192.168.174.133
 Transmission Control Protocol, Src Port: 21, Dst Port: 21
 File Transfer Protocol (FTP)
 [Current working directory:]

File Transfer Protocol (FTP): Protocol Packets: 646 - Displayed: 25 (3.9%) Profile: Default

```

220 (vsFTPd 3.0.5)
Name (192.168.174.133:kali): cyberarena
331 Please specify the password.
Password:
530 Login incorrect.
ftp> login failed
ftp>
ftp> put sample.txt
local: sample.txt remote: sample.txt
530 Please login with USER and PASS.
530 Please login with USER and PASS.
ftp> Can't bind for data connection: Address already in use
ftp>
  
```

FTP

Plaintext

Credentials visible

Readable commands

SFTP

Encrypted

Credentials hidden

Encrypted SSH packets

Week 5

```
cyberarena@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
cyberarena@ubuntu:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
cyberarena@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
cyberarena@ubuntu:~$ sudo ufw deny 80/tcp
Rule added
Rule added (v6)
cyberarena@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
80/tcp DENY IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) DENY IN Anywhere (v6)

cyberarena@ubuntu:~$
```

```
(kali@kali)~$
$ curl -I http://192.168.174.130

^C

(kali@kali)~$
$ ssh cyberarena@192.168.174.130
cyberarena@192.168.174.130's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-36-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat May  9 08:48:52 PM UTC 2026

System load:  0.15               Processes:    288
Usage of /:   80.1% of 38.09GB   Users logged in: 1
Memory usage: 56%              IPv4 address for ens33: 192.168.96.139
Swap usage:   13%

Expanded Security Maintenance for Applications is not enabled.

191 updates can be applied immediately.
137 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat May  9 20:39:10 2026 from 192.168.174.133
cyberarena@ubuntu:~$
```

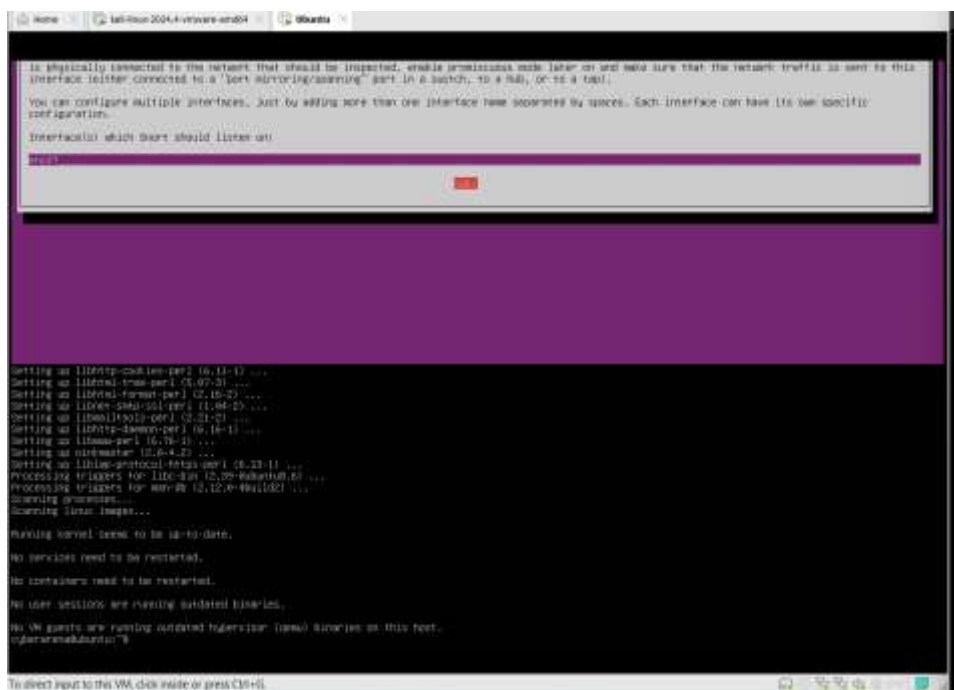
Commented [s6]: HTTP (curl) was blocked by UFW
SSH login still succeeded

Week 6

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ubuntu:~$ sudo apt-get -y install -o Dpkg::Options::=--no-install-recommends qemu-system-x86_64
05/05/24 05:56:27.77752 [==] [1:1439:11] QMP tcp tcp [==] [Classification: Attempted Information Leak] [Priority: 2] (TCP) 192.168.174.133:34892 -> 192.168.174.133:252
05/05/24 05:56:30.90664 [==] [1:1439:11] QMP tcp tcp [==] [Classification: Attempted Information Leak] [Priority: 2] (TCP) 192.168.174.133:34894 -> 192.168.174.133:252
05/05/24 05:56:30.93361 [==] [1:1439:11] QMP request tcp [==] [Classification: Attempted Information Leak] [Priority: 2] (TCP) 192.168.174.133:34896 -> 192.168.174.133:252
05/05/24 05:57:40.80201 [==] [1:1439:11] QMP request tcp [==] [Classification: Attempted Information Leak] [Priority: 2] (TCP) 192.168.174.133:34904 -> 192.168.174.133:252
05/05/24 05:57:43.93702 [==] [1:1439:11] QMP request tcp request [==] [Classification: Attempted Information Leak] [Priority: 2] (TCP) 192.168.174.133:34932 -> 192.168.174.133:252
05/05/24 05:57:59.9128 [==] [1:1439:11] QMP request tcp request [==] [Classification: Attempted Information Leak] [Priority: 2] (TCP) 192.168.174.133:34934 -> 192.168.174.133:252

```



```

(kali@kali)-[~]
$ sudo nmap -sS -p 1-1000 192.168.174.130
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 16:59 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00031s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 00:0C:29:EB:E6:BB (VMware)

Nmap done: 1 IP address (1 host up) scanned in 11.48 seconds
(kali@kali)-[~]
$

```

Week 8

```

Preparing to unpack .../1-libpcsc-lite1:2.0.3-1build1_amd64.deb ...
Unpacking libpcsc-lite:amd64 (2.0.3-1build1) ...
Selecting previously unselected package pcscd.
Preparing to unpack .../2-pcscd_2.0.3-1build1_amd64.deb ...
Unpacking pcscd (2.0.3-1build1) ...
Selecting previously unselected package libpcsclite1:amd64.
Preparing to unpack .../3-libpcsclite1_1.12.0-4ubuntu1~22.04.1build1_amd64.deb ...
Unpacking libpcsclite1:amd64 (1.12.0-4ubuntu1~22.04.1build1) ...
Selecting previously unselected package libpcsclite-helper1:amd64.
Preparing to unpack .../4-libpcsclite-helper1_1.12.0-4ubuntu1~22.04.1build1_amd64.deb ...
Unpacking libpcsclite-helper1:amd64 (1.12.0-4ubuntu1~22.04.1build1) ...
Selecting previously unselected package openssl-pkcs11:amd64.
Preparing to unpack .../5-openssl-pkcs11_0.25.0-7rc1-ubuntu0.21_amd64.deb ...
Unpacking openssl-pkcs11:amd64 (0.25.0-7rc1-ubuntu0.21) ...
Selecting previously unselected package openssl.
Preparing to unpack .../6-openssl_0.25.0-7rc1-ubuntu0.21_amd64.deb ...
Unpacking openssl (0.25.0-7rc1-ubuntu0.21) ...
Selecting previously unselected package openvpn.
Preparing to unpack .../7-openvpn_2.6.19-ubuntu0.24.04.1_amd64.deb ...
Unpacking openvpn (2.6.19-ubuntu0.24.04.1) ...
Selecting previously unselected package easy-rsa.
Preparing to unpack .../8-easy-rsa_3.1.7-2_all.deb ...
Unpacking easy-rsa (3.1.7-2) ...
Setting up libpcsc-lite1:amd64 (2.0.3-1build1) ...
Setting up libpcsclite1:amd64 (1.12.0-4ubuntu1~22.04.1build1) ...
Setting up libpcsclite-helper1:amd64 (1.12.0-4ubuntu1~22.04.1build1) ...
Setting up openssl-pkcs11:amd64 (0.25.0-7rc1-ubuntu0.21) ...
Setting up openssl (0.25.0-7rc1-ubuntu0.21) ...
Setting up openvpn (2.6.19-ubuntu0.24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/openvpn.service → /usr/lib/systemd/system/openvpn.service.
Setting up pcscd (2.0.3-1build1) ...
Created symlink /etc/systemd/system/sockets.target.wants/pcscd.socket → /usr/lib/systemd/system/pcscd.socket.
pcscd.service is a disabled or a static unit, not starting it.
Processing triggers for libc-bin (2.39-0ubuntu6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning Linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
cyberarena@ubuntu:~$

```

Commented [s7]: OpenVPN and Easy-RSA installed successfully.


```

cyberarena@ubuntu:~/openvpn-ca$ ./easyrsa sign-req server server
Using Easy-RSA 'vars' configuration:
* /home/cyberarena/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
You are about to sign the following certificate:
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.
Request subject, to be signed as a server certificate
for '825' days:

subject=
  commonName = server

Type the word 'yes' to continue, or any other input to abort.
Confirm request details: yes

Using configuration from /home/cyberarena/openvpn-ca/pki/openssl-easyrsa.cnf
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'server'
Certificate is to be certified until Aug 11 21:13:55 2025 GMT (825 days)

Write out database with 1 new entries
Database updated

Notice
-----
Certificate created at:
* /home/cyberarena/openvpn-ca/pki/issued/server.crt
cyberarena@ubuntu:~/openvpn-ca$

```

Commented [s10]: Sign Server Certificate

```

cyberarena@ubuntu:~/openvpn-ca$ ./easyrsa gen-req client client
Using Easy-RSA 'vars' configuration:
* /home/cyberarena/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank.
For some fields there will be a default value.
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [client]:client

Notice
-----
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /home/cyberarena/openvpn-ca/pki/reqs/client.req
* key: /home/cyberarena/openvpn-ca/pki/private/client.key
cyberarena@ubuntu:~/openvpn-ca$

```

Commented [s11]: Generate Client Certificate

```

cyberarena@ubuntu:/openvpn$ ./easyrsa sign-req client client1
Using Easy-RSA 'vars' configuration:
* /home/cyberarena/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
You are about to sign the following certificate:
Please check over the details shown below for accuracy. Note that this request
has not been cryptographically verified. Please be sure it came from a trusted
source or that you have verified the request checksum with the sender.
Request subject, to be signed as a client certificate
for '025' days:

subject=
commonName = client1

Type the word 'yes' to continue, or any other input to abort.
Confirm request details: yes

Using configuration from /home/cyberarena/openvpn-ca/pki/openssl-easyrsa.cnf
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName      :ASN.1 12:'client1'
Certificate is to be certified until Aug 11 21:21:13 2028 GMT (825 days)

Write out database with 1 new entries
Database updated

Notice
-----
Certificate created at:
* /home/cyberarena/openvpn-ca/pki/issued/client1.crt
cyberarena@ubuntu:/openvpn$ cat _

```

Commented [s12]: Sign Client Certificate

```

GNU nano 7.2 /etc/openvpn/server/server.conf
port 1194
proto udp
dev tun

ca /home/cyberarena/openvpn-ca/pki/ca.crt
cert /home/cyberarena/openvpn-ca/pki/issued/server.crt
key /home/cyberarena/openvpn-ca/pki/private/server.key

dh /home/cyberarena/openvpn-ca/pki/dh.pem

server 10.8.0.0 255.255.255.0

persist 10 120
persist-key
persist-tun

cipher AES-256-GCM
auth SHA256

user nobody
group nogroup

verb 3

help write Out where Is Cut Execute Location undo get Mark To be
Exit Read File Replace Paste Justify Go To Line redo Copy Copy Where

```



```
(kali@kali)-[~]
$ scp cyberarena@192.168.174.130:/home/cyberarena/client1.ovpn .
cyberarena@192.168.174.130's password:
client1.ovpn                                100% 1341    1.3MB/s   00:00
```

System information as of Sat May 8 10:01:08 PM UTC 2020

System load: 0.3 Processes: 1
Usage of /: 79.1% of 38.6MB Users logged in: 1
Memory usage: 63% [Pkg address for ensl]:
Swap usage: 3% [Pkg address for ensl]:

Expanded Security Maintenance for Applications is not enabled.
191 updates can be applied immediately.
137 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>.

Last login: Sat May 8 10:48:51 2020 from 192.168.174.133
cyberarena@kali:~\$ sudo ufw allow 1194/tcp
[sudo] password for cyberarena:
Rule added
Rule enabled (v6)
cyberarena@kali:~\$ sudo ufw status
Status: active

To	Action	From
22/tcp	ALLOW	Anywhere
80/tcp	DENY	Anywhere
Anywhere on vxlan.calico	ALLOW	Anywhere
Anywhere on cali+	ALLOW	Anywhere
1194/tcp	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)
80/tcp (v6)	DENY	Anywhere (v6)
Anywhere (v6) on vxlan.calico	ALLOW	Anywhere (v6)
Anywhere (v6) on cali+	ALLOW	Anywhere (v6)
1194/tcp (v6)	ALLOW	Anywhere (v6)
Anywhere	ALLOW OUT	Anywhere on vxlan.cal
Anywhere	ALLOW OUT	Anywhere on cali+
Anywhere (v6)	ALLOW OUT	Anywhere (v6) on vxl
Anywhere (v6)	ALLOW OUT	Anywhere (v6) on cal

Commented [s13]: Kali connected to the OpenVPN server, encrypted tunnel established, tun0 interface created, VPN IPs assigned successfully

Capturing from eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

udp.port == 1194

No.	Time	Source	Destination	Protocol	Length	Info
8822	6299.7811284	192.168.174.130	192.168.174.133	OpenVPN	82 Mes	
8825	6306.0415444	192.168.174.133	192.168.174.130	OpenVPN	82 Mes	
8826	6308.9478589	192.168.174.130	192.168.174.133	OpenVPN	82 Mes	
8827	6311.3270263	192.168.174.133	192.168.174.130	OpenVPN	114 Mes	
8828	6319.2077802	192.168.174.130	192.168.174.133	OpenVPN	82 Mes	
8829	6321.4365546	192.168.174.133	192.168.174.130	OpenVPN	82 Mes	
8830	6329.5603520	192.168.174.130	192.168.174.133	OpenVPN	82 Mes	
8832	6331.7380897	192.168.174.133	192.168.174.130	OpenVPN	82 Mes	
8835	6339.5784398	192.168.174.130	192.168.174.133	OpenVPN	82 Mes	
8836	6342.6207116	192.168.174.133	192.168.174.130	OpenVPN	82 Mes	

Frame 7466: 56 bytes on wire (448 bits) @ 0.000 00 0c 29 eb e6 bb 00 0c 29 69 01
Ethernet II, Src: VMware 69:01:4f (08:00:06:09:10:4f), Dst: 08:00:06:09:10:4f
Internet Protocol Version 4, Src: 192.168.174.133, Dst: 192.168.174.130
User Datagram Protocol, Src Port: 34431, Dst Port: 1194
OpenVPN Protocol

wireshark_eth1XWUO3.pcapng Packets: 8836 - Displayed: 84 (1.0%) Profile: Default

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

udp.port == 1194

No.	Time	Source	Destination	Protocol	Length	Info
8880	6431.6388881	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8881	6432.0628088	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8882	6433.0889906	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8883	6434.1116714	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8884	6435.1349537	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8885	6436.1598191	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8886	6437.5535685	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8887	6438.5634022	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8888	6439.5867832	192.168.174.133	192.168.174.130	OpenVPN	150	Mes
8889	6439.5876392	192.168.174.130	192.168.174.133	OpenVPN	82	Mes

Frame 7486: 56 bytes on wire (448 bits) 0000 00 0c 29 eb e6 bb 00 0c 29 09 01
 Ethernet II, Src: VMware_09:01:4f (00:0c:09:01:4f:00:00) 00 2a 38 8b 49 08 48 11 23 0f c6
 Internet Protocol Version 4, Src: 192.168.174.133, Destination: 192.168.174.130 ac 82 86 7f 04 aa 08 16 dc 88 38
 User Datagram Protocol, Src Port: 34431, Destination Port: 1194 a0 e2 a9 00 00 00 00 00
 OpenVPN Protocol

File Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history

kali@kali:~\$

~\$ ping 10.0.0.5

PING 10.0.0.5 (10.0.0.5) 56(84) bytes of data:

64 bytes from 10.0.0.5: icmp_seq=1 ttl=64 time=0.054 ms

64 bytes from 10.0.0.5: icmp_seq=2 ttl=64 time=0.054 ms

64 bytes from 10.0.0.5: icmp_seq=3 ttl=64 time=0.054 ms

64 bytes from 10.0.0.5: icmp_seq=4 ttl=64 time=0.054 ms

64 bytes from 10.0.0.5: icmp_seq=5 ttl=64 time=0.054 ms

5 ping requests: 0% packet loss, time 0.10s

wireshark_eth1AXWUC

CP ALLOW

CP DENY

tcp on vxlan.calico ALLOW

tcp on cali+ ALLOW

udp ALLOW

ip (v6) ALLOW

ip (v6) DENY

Week 9

```
(kali㉿kali)-[~]
$ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519
Generating public/private ed25519 key pair.
Enter passphrase for "/home/kali/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Passphrases do not match. Try again.
Enter passphrase for "/home/kali/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kali/.ssh/id_ed25519
Your public key has been saved in /home/kali/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:PRwUcYLdJXJbZwnkTzRcAXWPgkn42JY+rx30Tgb65Iw kali@kali
The key's randomart image is:
+--[ED25519 256]--+
|                 |
|   oo.B+==B=0|   |
|   ++.0....|   |
|   .0=..0|   |
|   So+ .|   |
|   o.|   |
|   0*|   |
|   #..|   |
|   Eo0|   |
+--[SHA256]-----+
(kali㉿kali)-[~]
$
```

Commented [s14]: Generate SSH Key on Kali

```
(kali㉿kali)-[~]
$ ssh-copy-id -i ~/.ssh/id_ed25519.pub cyberarena@192.168.174.130
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kali/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
cyberarena@192.168.174.130's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/kali/.ssh/id_ed25519 cyberarena@192.168.174.130"
and check to make sure that only the key(s) you wanted were added.
```


now is logged without password

```
hal@kali:~$ ssh cyberarena@192.168.174.130
Warning: Permanently added '192.168.174.130' (SSH-2.0-ubuntu_openssh_8.9p1)
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 5.14.0-36-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat May  9 10:15:49 PM UTC 2026

System load:  0.17               Processes:    293
Usage of /:   79.1% of 38.09GB   Users logged in: 1
Memory usage: 65%              IPv4 address for ens33: 192.168.86.139
Swap usage:   3%

Expanded Security Maintenance for Applications is not enabled.

191 updates can be applied immediately.
137 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Sat May  9 22:01:50 2026 from 192.168.174.133
cyberarena@ubuntu:~$
```

```
vim /etc/ssh/sshd_config
This is the sshd server system-wide configuration file. See
sshd_config(5) for more information.

This sshd was compiled with PATH=/usr/local/sbin:/usr/sbin:/usr/bin:/usr/games

The strategy used for options in the default sshd_config shipped with
OpenSSH is to specify options with their default values where
possible, but leave them commented. Uncommented options override the
default value.

Include /etc/ssh/sshd_config.d/*.conf

When system socket activation is used (the default), the socket
configuration must be re-generated after changing Port, AddressFamily, or
ListenAddress.

For changes to take effect, run:

  systemctl daemon-reload
  systemctl restart ssh.socket

#Port 22
AddressFamily any
ListenAddress 0.0.0.0
ListenAddress ::

HostKey /etc/ssh/ssh_host_rsa_key
HostKey /etc/ssh/ssh_host_ecdsa_key
HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyInterval 0
#RekeyLimit 0

# Logging
SyslogFacility AUTH
LogLevel INFO

# Authentication:

#LoginGraceTime 3m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

vim
```

Change to

```
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication no
#PermitEmptyPasswords no
```

```
cyberarena@ubuntu:~$ sudo systemctl reload apache2
cyberarena@ubuntu:~$
```

```
kali@kali:~$
kali@kali:~$ nmap -sV 192.168.174.130
[info] password for nmap:
Starting Nmap 7.95SN ( https://nmap.org ) at 2020-05-09 18:45 EDT
Nmap scan report for 192.168.174.130
Host is up (6.00007s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.8p1 Ubuntu 3ubuntu1.14 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 256 af:ca:4e:53:0b:12:ce:73:da:9e:fece:b9c:70:9e:ca8 (ECDSA)
|_ 256 9d:82:aa:12:ee:66:fd:5d:96:9e:ff:56:7c:08:05:83 (ED25519)
8080/tcp   open  http     Apache/2.4.18
|_ http-title: HTTP Status 404 (Not Found)
MAC Address: 98:8C:20:EB:1A:8E (VMware)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose/storage-misc
Running (JOST GMSSTAG): Linux 4.X|5.X|2.6.X|3.X (97%), Synology DiskStation Manager 5.X (99%), Netgear R6700v2 4.X (97%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:3 cpe:/o:synology:d
iskstation_manager:5.2 cpe:/o:netgear:r6700v2:4.2.36
Aggressive OS guesses: Linux 4.15 - 5.8 (97%), Linux 5.0 - 5.4 (97%), Linux 5.0 - 5.5 (95%), Linux 5.4 (91%), Linux 2.6.32 (91%), Linux 3.10 -
4.9 (91%), Linux 3.2 - 4.9 (91%), Linux 3.4 - 3.10 (91%), Linux 5.1 (91%), Linux 2.6.32 - 3.10 (91%)
No exact OS matches for host (test condition non-ideal).
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
Hop RTT      Address
1  0.47 ms  192.168.174.130

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 26.97 seconds
kali@kali:~$
```